

Final Report

Team 25

GenAI Housing Prices Prediction

Groupmates:

CHENG Kit Ling, 21167863

FU Chung Yan, 21241750

LAM Madison Emily, 21333591

YANG Yulai, 21220811

YANG Zimo, 20936815

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dataset:https://drive.google.com/file/d/1kzvwS09YwZ2F5muONBb4u0seua9_qtAa/view?usp=sharing

https://drive.google.com/file/d/1XXh3Rhu3_oCq-in5QIb-wFvHjX8VMDCA/view?usp=sharing

Problem Definition

Nowadays, manual evaluation is still the major method that is adopted in the traditional real estate appraisal industry. However, the approach is time-consuming, lacking transparency (Dong Ruilin, Dong Nan; 2019). In Hong Kong, traditional property valuation adopts a cost, income, sale comparison approach. It calculates the value of a property, including the factors such as replacement cost, less depreciation plus the market-derived land value, thereby demonstrating the method in a complicated way (Deng & Zhang, 2025).

Our team also observed that sellers and buyers need to spend so much time on clicking categories like districts, range of prices, types of estate on property websites for finding the transaction prices of the similar properties. Sometimes, the results show different transaction prices. It is hard for the sellers to predict the accurate prices of their properties.

Based on the above challenges, the advocacy of Machine learning is becoming increasingly prevalent in predicting the property valuation in different countries in order to handle the increasing transaction volume of properties. In machine learning, big data collect information, Method-oriented and Features-oriented studies could demonstrate the advanced prediction accuracy (Deng & Zhang, 2025).

Our team decided to create a GAI-tool which will collect different types of factors and an enormous dataset which influence the prediction for transaction prices of the properties, further estimating the potential transaction prices directly for sellers and buyers based on the past transactions' record . They could chat with AI agents about the housing price predictions

at any time and anywhere, breaking the geographical and spatial limitations, enabling clients to obtain the precise transaction prices for their own properties in a more convenient and efficient way.

Data Collection & Preprocessing

The dataset used is available via Kaggle under the title “Hong Kong Housing Price (2020-2023)” (CyrusTTF, 2020). This dataset is suitable for the project’s goal of predicting house prices because it considers a wide range of explanatory variables, with approximately 149,882 records. The robustness of data boosts the statistical power of our model.

The data samples are from a relevant time frame which allows us to train our model on up-to-date market trends. Furthermore, the dataset classifies housing data on the district level, which is essential in a location-driven Hong Kong market. Another notable feature we noticed in the changes feature, which refers to the percentage change in price since the last time this specific unit was sold. A high positive number in this category may be a strong indicator that the property has good resale investment value. Lastly, the temporal feature allows us to analyze potential seasonal trends and differences.

Preprocessing is completed in KNIME where the workflow is illustrated in Figure 1. First, we use a CSV Reader to load and inspect the data. To encode categorical data and ensure consistent handling of all features, One to Many is used to encode District. We go on to parse the date feature with String to Date&Time and Date&Time Extractor, creating new numerical columns Year and Month. String manipulation is used to handle the percentage sign in the changes feature and missing values are replaced with 0 as a catch-all for stable or missing prior data. Since real estate prices are usually skewed, with cheap housing as the majority, a normalizer is applied to ensure fair representation of data.

In conclusion, our pre-processing focused on feature engineering, specifically extracting hidden signals from texts and converting them into numerical values that a machine learning model can understand. Having structured data provides the foundation for training the regression and KNN models that will be explained in the next section.

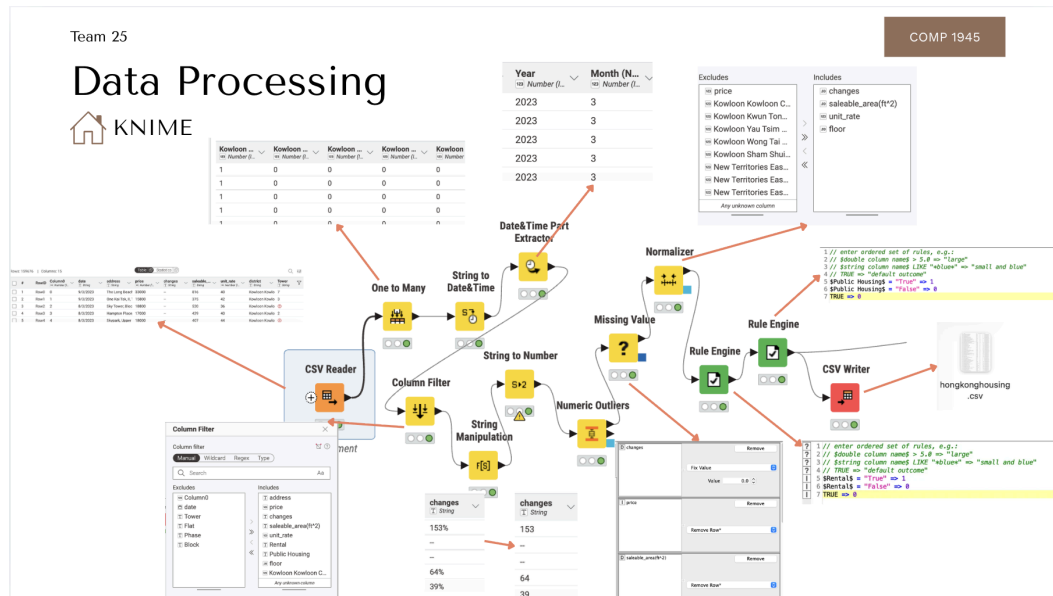


Figure 1: Diagram showing pre-processing workflow, completed in KNIME

System and Model Design

End-user interaction with the proposed real estate AI agent follows a natural-language-driven workflow, starting with an unstructured prompt where users articulate their real estate needs without explicitly declaring their role. Powered by a Large Language Model, the agent autonomously infers whether the user is a buyer or seller by analyzing input semantics—identifying buyer intent through terms like “looking for a flat” or “budget constraints,” and seller intent via mentions of “listing a property” or “asking price.” The LLM then parses the input to extract key structured information and proactively asks follow-up questions to fill gaps, ensuring all data required for downstream tasks is complete.

This conversational interface eliminates the need for complex forms, making the solution accessible to users of all technical literacy levels.

Once the user's role and requirements are confirmed, the LLM orchestrates role-specific parallel execution pipelines to optimize efficiency. For buyers, the LLM concurrently queries Hong Kong's major real estate databases—including Centaline Property, Midland Realty, and Squarefoot's public APIs, as well as the Hong Kong Housing Authority's public housing datasets—to retrieve geospatially filtered listings, while invoking a fine-tuned price prediction model for data-driven valuations. For sellers, parallel tasks include generating platform-specific channel recommendations based on market reach and audience demographics, creating tailored marketing copy, and conducting competitor analysis by scraping and benchmarking comparable district listings. The LLM synthesizes these results into actionable deliverables including curated listings with price estimates for buyers and comprehensive marketing plans for sellers. It then presents the outputs to users and enables a seamless handoff to a human agent when necessary.

The core function of the AI agent is to streamline real estate transactions via the LLM's central orchestration. As the hub of the system, the LLM translates unstructured user needs into structured actions, coordinates parallel data retrieval and analysis to replace manual, fragmented operations, and delivers role-specific, actionable outputs. It bridges the gap between users' natural-language requests and the structured support required for real estate transactions, reducing user effort and improving efficiency without redundant steps.

AI Agent Workflow

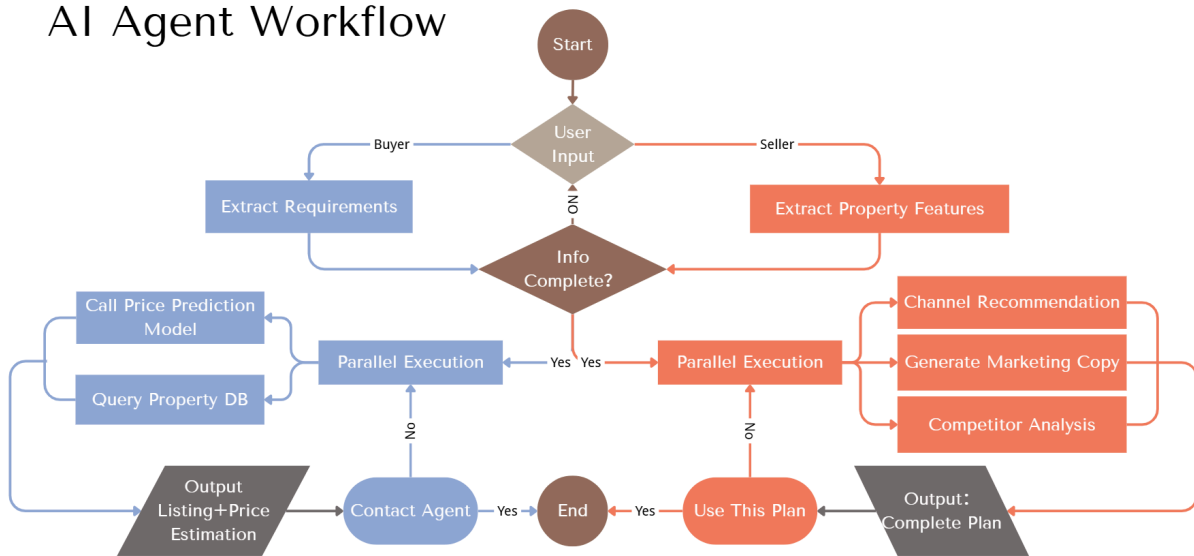


Figure 2: AI Agent Workflow as Buyer and Seller

KNN and Linear Regression Comparison

For our case housing prices in Hong Kong, we worked with KNN and linear regression. Like most models, both methods had their advantages and disadvantages. For starters, KNN was easier to explain to users, worked directly on the database provided by Kaggle without any training required and did not assume any certain relationship. But, while being locally adaptable, KNN was heavily dependent on the data, computationally slow and did not learn on a global model.

On the other hand, while the linear regression model in our case was slow to adapt to the ever changing housing market since it required retraining, it was computationally faster compared to KNN after initial training. Additionally, it describes a global relationship between feature and price, capturing the trends with smooth, continuous relationships.

In short, both models were suitable for our problem in real estate for users in Hong Kong. We utilized the strengths of each to come up with a coherent solution in the form of a chatbot.

Future Ideas

While we are currently using a KNN model for housing price predictions in Hong Kong, a future idea is to use a convolutional neural network, or CNN. With CNN, we would be able to capture the visual quality of the house which includes complex non-linear patterns and therefore, better generalizations when comparables are sparse. But, CNN requires a large labeled dataset with images and transaction price and is less transparent to users.

Furthermore, since CNN is computationally more expensive compared to KNN and Linear Regression, it would require additional resources to execute.

Nonetheless, there would be two main applications of CNN with our Gen-AI housing chatbot. The first is an image-based valuation where we would provide photos of each house with multiple views into a CNN. This would allow the network to learn the unique visual features, including renovation quality, lighting, sea view and building age. Then, the output would be the premium or discount factor relative to baseline transaction prices.

Next, the second utilization is with a hybrid CNN and structured data. This would produce structured outputs of the estate name, floor, net area and transaction history. The regression layer would be able to predict property price per area, capturing the non-linear interactions between visual cues and numerical attributes.

Overall, both uses of a convolutional neural network would be able to take in the visual qualities of each house and produce a complex, non-linear output that the KNN and regression models were not able to.

Evaluation & Deployment Plan

To assess the AI agent's success, we adopt a mixed-method evaluation framework that combines quantitative system performance metrics and qualitative user feedback.

Quantitatively, we track three key metrics to measure the agent's operational effectiveness: the button clicking rate, which gauges how often users act on the agent's recommendations (such as selecting "Use This Plan" or "Contact Agent") to validate alignment with user intent; the average dialogue turn count, where fewer turns signal more efficient information gathering and reduced user effort; and sampling audits of interaction logs to verify the accuracy of the LLM's role classification, requirement extraction, and parallel task execution. Complementing these metrics, qualitative feedback is collected through two channels: satisfaction rate surveys to measure perceived usability and helpfulness, and open-ended feedback to capture nuanced pain points and improvement suggestions that quantitative data may overlook.

For real-world deployment, we implement a phased rollout strategy: first, a closed beta with a small user group validates the end-to-end workflow, identifies edge cases, and refines the model based on initial user data; following successful testing, the agent is gradually integrated into existing digital channels, including website chatbots and mobile app interfaces, with continuous monitoring of the above evaluation metrics to support iterative fine-tuning and ensure the agent adapts to evolving user needs and market conditions over time.

Risks & Limitations

What are the potential risks and limitations of our housing-price predicting AI agent? Firstly, there might be some hallucinations. Since our agent is built on a Large Language Model, there is no absolute guarantee that for every answer the properties or prices are grounded in reality. To mitigate this issue, we implemented strict prompt engineering, explicitly instructing the LLM that 'All price figures must come from actual transaction

records in the knowledge base. Never fabricate any transaction data.’ Secondly, the users might over rely on the agent’s answers without understanding its limitations. To address this, we designated the chatbot to add a disclaimer. If the knowledge base returns no relevant results, the agent will honestly state that insufficient comparable data is available and suggest the user consult a professional surveyor.

Thirdly, the data is static. Since our current knowledge is built on a CSV file, it does not reflect real-time market fluctuations or new listings. Consequently, users might receive outdated information, leading to suboptimal decisions. We plan to solve this problem by manually updating the CSV files and re-uploading them to the knowledge base on a weekly or monthly basis to ensure temporal relevance. Lastly, the chatbot might have performance issues under heavy traffic, like latency issues or crashes. So if traffic actually increases, we plan to monitor performance metrics like response time while testing the system under simulated high traffic. If an issue occurs in the test, we can fine-tune the settings in Dify or upgrade server capacity. By addressing the above mentioned risks, we ensure our AI agent remains a reliable and trustworthy assistant.

AI Usage Statement

In this project, we utilized AI tools and platforms to transform our conceptual design into an operational prototype. We developed the Gen-AI housing price prediction chatbot through the Dify platform. We designed a crafted system prompt which is further optimized by AI that controls AI agent’s behavior according to the questions of clients; Hong Kong historical property transaction records for similarity-based retrieval, and a conversational interface that supports both text and speech-to-text input. The LLM model provides buyers

with price trend insights while offering sellers comparable listing analyses that influence their pricing strategies, solving their questions specifically.

Bibliography

董睿琳, & 董楠. (2019). 基于房地产大数据的自动估价系统研究. 智能计算机与应用.

CyrusTTF. (2020). *Hong Kong Housing Price (2020-2023)*. [Kaggle.com](https://www.kaggle.com/datasets/cyrusttf/hong-kong-housing-price-2020-2023).

<https://www.kaggle.com/datasets/cyrusttf/hong-kong-housing-price-2020-2023>

Deng, L., & Zhang, X. (2025). Boosting the accuracy of property valuation with ensemble learning and explainable artificial intelligence: The case of Hong Kong. *The Annals of*

Regional Science, 74(1), Article 32. <https://doi.org/10.1007/s00168-025-01365-7>

Alashqar, Belal. "EDA + Data Cleaning (Ames Housing Dataset)." Kaggle, 2021,

www.kaggle.com/code/abdelhai/eda-data-cleaning-ames-housing-dataset.